

LECTURE PLAN – COMPUTER ARCHITECTURE

Lecture 33 of 45

Topic: Pipelining

Unit: Unit 6

Learning Outcomes (LO)

- LO1: Define and explain the basic concepts of Pipelining. (L1 – Remember)
- LO2: Explain the working and important features of Pipelining. (L2 – Understand)
- LO3: Apply the concept of Pipelining to simple examples or problems. (L3 – Apply)

Teaching Content

- Pipeline concept
- Pipeline stages
- Throughput and speedup idea

Teaching–Learning Activities

Activity	Description
Demo	Teacher explains Pipelining using simple examples on the board/PPT.
Peer Activity	Students discuss a short example in pairs and compare their answers.
Board Problem	Students solve one basic problem based on Pipelining with teacher guidance.

Assessment Methods

- Quiz: Check understanding of key terms and concepts.
- Assignment: Solve/write a short problem related to the lecture topic.
- Viva: Explain the main concept and one example in simple words.

Teaching Aids

- PPT with examples, whiteboard, textbook diagrams, and simple classroom examples.